

A Correlational Study Between Psychological Skills and Competitive Anxiety Among National Level Sprinters

Nita Pal*, Dr. Deepak Mishra**, Dr. Nilima Deshpande***

ABSTRACT

Competition can cause athletes to react both physically (somatic) and mentally (cognitive) in a manner which can affect their performance abilities. The purpose of the present study was to analyze the competitive anxiety and psychological skills of the national level sprinters and thereby find out the correlation between those parameters. For this study 30 sprinters were selected. All selected athletes had participated at national level. The age of the subjects was ranged between 18 to 25 years.

The data were collected using standardized questionnaires, i.e., Competitive State Anxiety Inventory-2 (CSAI-2) and Psychological Performance Inventory (PPI). Mean and standard deviations of the variables were calculated. Pearson correlation coefficient was also used to find out the relationship among selected parameters. It is found that the mean value of cognitive and somatic state anxiety of the athletes is low whereas the mean value of self-confidence of the athletes is high though it has a scope for improvement as the mean value of self-confidence is slightly higher than the moderate zone. The study also indicates that there is scope for improvement of psychological skills especially for the negative energy, attention control, visual & imagery control and positive energy of the athletes. On the other hand, the athletes possess excellent skills on self-concept, motivational level and attitude control. A negative correlation between cognitive state anxiety and psychological skills has been found which indicates that increment of cognitive state anxiety decreases the psychological skills and also decrement of the former enhances the latter. Out of seven parameters of psychological skills only self-concept has been found to be significantly correlated with cognitive state anxiety. Psychological skills also have been found to be negatively correlated with somatic state anxiety which implies increment of one decreases the other. The parameters Self- concept, negative energy, attention control and attitude control are having statistically significant correlations with somatic state anxiety level of the athletes. Psychological skills of the athletes are more influenced by the somatic state anxiety rather than the cognitive state anxiety thereby more focus should be given to control the somatic anxiety level of the athletes. Self-confidence is positively correlated with all the parameters of psychological skills but no significant relations have been found between self-confidence and the parameters of psychological skills.

Key words: Somatic Anxiety, Cognitive Anxiety, Self-confidence, Psychological skill

* Master in Sports Coaching Trainee, email- siddhanita94@gmail.com

** SO, Department of Sports Psychology, SAI NSNIS Patiala

*** Coach, Athletics, SAI NSNIS Patiala

INTRODUCTION

Sports is a complex phenomenon which requires not only the physical fitness of athletes but psychological fitness is equally important to excel in the increasingly growing sports competitions. The aim of sports psychology is to understand the contribution that psychological components have on exercise performance, but also the effects that participation in physical exercise has on psychological development. Sports psychologists often talk about the significance of being in the “zone” and of creating an intimidating presence for the opponent on the field of play, but waiting until the player is in competition to apply the principles of sports psychology, without previous training will not allow for an enhanced performance. So, before this we need to apply a winning attitude and the ability to apply it to our training, which is one of the most vital and difficult areas to apply it. The psychological fitness completes the total spectrum of fitness required for executing psychological skill and achieving peak level performance in sports. *In his book Mental Toughness Training for Sports: Achieving Athletic Excellence* (1986) James Loehr characterizes individuals that are mentally fit as:

- Self-motivated & self-directed
- Positive but realistic
- Calm & relaxed under fire
- Highly energetic and ready for

action

- Determined
- Mentally alert & focused
- Self-confident
- Fully responsible

Competition can cause athletes to react both physically (somatic) and mentally (cognitive) in a manner which can affect their performance abilities. Stress, arousal and anxiety are terms used to describe condition. "Anxiety is the negative aspect of experiencing the stress. It is the worry that is experienced due to fear of failure." (Honey Bourne, 2003). Accordingly, there are two types of state anxiety. The first one, cognitive state anxiety, is a feeling of nervousness, apprehension, worry thoughts, loss of concentration that a performer has about their lack of ability to complete a task (cognitive + psychology). The second, somatic state anxiety is a physiological response to the situation or awareness of arousal (Increased heart rate, sweaty palms, Muscle tension) in body where the performer feels they may not cope. (Somatic = physiological) (Burton, 1988). Worrying about how an athlete is going to perform also leads to anxiety about his performance. According to Vanthuyne (1999) we don't worry just in our heads, our entire body worries resulting in a body reaction as well as a mental reaction, this fact is critical regarding athletic performance. Due to its objective nature, athletics can often be a frustrating and stressful sport. In order to

combat increased anxiety levels, athletes can practice mental skills and strategies to help them relax on and off the field. Using relaxation strategies can help control anxiety and is important in enhancing performance. Thus, a challenge for an individual performer is to attain and maintain a psychological state that facilitates performance, given his or her personality, the specific task, and the wider task environment. The ability to attain such a state is thought to depend, in part, on the use of psychological skills. Such skills are thought to be learnable.

Psychological skills comprise the deliberate use of pre-prepared and structured sequences of specific thoughts and behaviors by athletes and exercisers to regulate their psychological state (e.g., feelings of confidence). Psychological skills involve skill of stress management, conflict reduction, emotional regulation, attention and goal setting skill and also leadership and communication skill. Psychological skills are of interest within sport and exercise psychology (SEP) because psychological states are thought to affect performance in sport and exercise settings and, thus, if performers are able to regulate their psychological states, they may be able to enhance their performance. Within current theory, it is proposed that a performer's psychological state (e.g., self-confidence, motivation, anxiety, and attention) is considered to affect his or her performance. The nature of the

relationship between a performer's psychological state and his or her performance depends on three factors: the individual performer, the specific task at hand, and the wider task environment. A lot of research has been done to study the role of psychological skills in developing excellence in athletes and other sportspersons. Here, the emphasis is laid on goal-setting, mental imagery, relaxation, and self-talk skills.

The present study attempts to find out the psychological skills and level of anxiety of the national level sprinters and ultimately find out the relationship between the parameters of those.

METHOD

Sample

The sample consists of thirty (N=30) subjects who are national level sprinters. The purpose of the present study was to analyze the psychological skills and competitive anxiety of the subjects. The age range of the subjects was between 18 to 25 years. The subjects selected for the study were both males and females.

Tools

1. The competitive anxiety of the athletes was calculated numerically by CSAI-2 test. The Competitive State Anxiety Inventory-2 (CSAI-2), developed by Martens, Vealey, and Burton (1990), is one of the most frequently used instruments when assessing competitive state anxiety in sport psychology research. It is a multidimensional test of competitive A

state, which is situation-specific, and accurately assesses three different dimensions or manifestations of state anxiety: cognitive state anxiety, somatic state anxiety, and self-confidence state anxiety.

A score between 9 to 18 indicates a low state anxiety, that is between 19 to 27 indicates moderate state anxiety and that is between 28 to 36 indicates a high state anxiety.

2. The psychological skills of the athletes were measured through the Psychological Performance Inventory (PPI) test. The PPI was developed by Loehr in 1986. The PPI is a self-report scale, rather than interview-based, assessment.

A score between 6 to 18 indicates a

need for special attention (NSA), that between 19 to 25 indicates room for improvement (RFI) and that between 26 to 30 indicates excellent skills (ES).

Statistical Approach

1. The data obtained were compiled and statistically analyzed with the help of SPSS software.

2. The mean and standard deviation were calculated.

3. The Pearson correlation coefficients were calculated to find out the correlation between the variables.

RESULTS AND DISCUSSIONS

After analyzing the data using the SPSS software the following results were obtained which are depicted below in tabular form.

Table-1: Socio Demographic Characteristics of the Sample

	M ± S. D
Age	23.56 ± 1.45
Experience	5.22 ± 1.95

Table 1 shows the age and experience distribution of the sample. The mean age of the participants was 23.56± 1.45 while

the mean game experience of the participants was 5.22± 1.95.

Table-2: Competitive Anxiety of the Sample

Variables	M± S. D	Remark
Cognitive State Anxiety	16.03±4.29	Low
Somatic State Anxiety	16.17±3.53	Low
Self Confidence	28.06±2.61	High

The results represented in table 2 show the mean and standard deviation of the cognitive state anxiety, somatic state anxiety and self-confidence level of the

participants. The values indicate that the participants have low level of cognitive and somatic state anxiety while they have high level of self-confidence level.

Table-3: Psychological Skills of the Participants

Variables	M± S. D	Remark
Self Confidence	26.70±2.41	Excellent Skill
Negative Energy	23.26±2.61	Room for Improvement
Attention Control	23.20±2.20	Room for Improvement
Visual & Imagery Control	25.46±2.48	Room for Improvement
Motivational Level	28.17±1.49	Excellent Skill
Positive Energy	25.90±2.06	Room for Improvement
Attitude Control	27.10±2.11	Excellent Skill

Table 3 shows the distribution of the parameters of psychological skills. Data indicates that in the field of negative energy, attention control, positive energy

and visual and imagery control there is room for improvement (RFI). For other parameters they have excellent skills (ES).

Table-4: Correlation Between Cognitive State Anxiety and Psychological Skills of the Sample

Variables	Pearson Correlation Coefficient	Significance Level
Cognitive State Anxiety	1	
Self Confidence	-.416*	.022
Negative Energy	-.105	.580
Attention Control	-.143	.451
Visual & Imagery Control	-.185	.326
Motivational Level	-.125	.510
Positive Energy	-.109	.567
Attitude Control	-.301	.105

**Significant at .01 level * Significant at .05 level

The results presented in the table-4 shows that all the parameters of psychological skills have statistically negative correlation with cognitive state anxiety. It also shows that cognitive state anxiety has a statistically significant relationship

with self-confidence ($r = -.416$) at .05 level. The researcher again found that the correlation between cognitive state anxiety and the other parameters of psychological skills such as negative energy, attention control, visual and

imagery control, positive energy, motivational level and attitude control was statistically insignificant. The

minimal relationship was found between negative energy and cognitive state anxiety ($r = -.101$).

Table-5: Correlation Between Somatic State Anxiety and Psychological Skills of the Sample

Variables	Pearson Correlation Coefficient	Significance Level
Somatic State Anxiety	1	
Self Confidence	-.575**	.001
Negative Energy	-.440*	.015
Attention Control	-.746**	.000
Visual & Imagery Control	-.123	.515
Motivational Level	-.282	.131
Positive Energy	-.335	.070
Attitude Control	-.448*	.013

**Significant at .01 level * Significant at .05 level

Table 5 presents the results of correlation between the somatic state anxiety and all the parameters of psychological skills. Somatic state anxiety has statistically negative correlation with all the parameters of psychological skills. Results also reveal that somatic state anxiety has statistically significant relationship with self-confidence ($r = -.575$) at .01 level. Significant relationship was also found between somatic state anxiety and negative energy ($r = -.440$) at .05 level. The relationship between

somatic state anxiety and attention control is statistically very significant ($r = -.746$) at .01 level. There is also significant relationship between somatic state anxiety and attitude control ($r = -.448$) at .05 level. The relationship between somatic state anxiety and visual and imagery control, motivational level and positive energy is statistically insignificant. The minimal relationship was found between visual and imagery control and somatic state anxiety ($r = -.123$).

Table-6: Correlation Between Self Confidence and Psychological Skills of the Sample

Variables	Pearson Correlation Coefficient	Significance Level
Self-confidence	1	
Self Confidence	.222	.237
Negative Energy	.240	.202
Attention Control	.339	.067
Visual & Imagery Control	.202	.284
Motivational Level	.015	.938
Positive Energy	.297	.112
Attitude Control	.093	.626

**Significant at .01 level * Significant at .05 level

The results of table 6 show that self-confidence is positively correlated with all the parameters of psychological skills

but no significant relations have been found between self-confidence and the parameters of psychological skills.

REFERENCES

- 1 **Aufenanger, S. J., (2005).** Relationship between mental skills and competitive anxiety interpretation in open skill and closed skill athletes.
http://rave.ohiolink.edu/etdc/view?acc_num=miami1117130981
- 2 **Baćanac, L., Milićević-Marinković, B., Kasum, G. & Marinković, M. (2014).** Competitive anxiety, self-confidence and psychological skills in top athletes with and without disabilities: Pilot study. *Physical Education and Sports, Vol. 12, No 2, 59-70.*
- 3 **Beltramo, M. (2020).** National level sprinter's competitive anxiety and performance success according to ability level and sex: an observational study with a cross-sectional design.
- 4 **Chakarvarti, D. & Lal, M. (2016).** Emotional Intelligence and Its Association with Social Physique Anxiety and Performance among Sprinters. *Rupkatha Journal on Interdisciplinary Studies in Humanities* (ISSN 0975- 2935) Special Issue, Vol. VIII, No. 2.
- 5 **Dockrat, S. Y. (2006).** The study of the impact of a psychological skills training programme on the performance of sprinters. <http://hdl.handle.net/2263/28299>

- 6 **Fletcher, D. & Hanton, S. (2001).** The relationship between psychological skills usage and competitive anxiety responses. *Psychology of Sports and Exercise*, Vol-2, Issue-2, 89-101.
- 7 **Hagan, J. E., Pollmann, D. & Schack, T. (2017).** Elite athletes' in- event competitive anxiety responses and psychological skills usages under different condition. <https://doi.org/10.3389/fpsyg.2017.02280>
- 8 **Hannibalsson, K.A. (2018).** Psychological skills, mental toughness, anxiety and performance in young basketball players. <http://hdl.handle.net/1946/31273>
- 9 **Hanton, S., Cropley, B., Neil, R., Mellalieu, S. D. & Miles, A. (2007).** Experience in sports and its relationship with competitive anxiety.” *International Journal of Sport and Exercise Psychology*, Vol. 5, Issue 1, 28-53.
- 10 **Hanton, S., Wadey, R. & Connaughton, D. (2005).** Debilitative interpretations of competitive anxiety: A qualitative examination of elite performers. *European Journal of Sports Science*, Vol. 5, Issue 3, 123- 136.
- 11 **Hayslip, B., Petrie, T. A., MacIntire, M. M. & Jones, G. M. (2010).** The influence of skill level anxiety and psychological skills use in amateur golfers' performance. *Journal of Applied Sports Psychology*, Vol. 22, Issue 2, 123-133.
- 12 **Jones, L. & Stuth, G. (1997).** The use of mental imagery in athletics: An overview. *Applied and Preventive Psychology*, Vol-6, Issue-2, 101-115.
- 13 **Jones, M. & Uphill, M. (2004).** Responses to the competitive state anxiety inventory-2 (d) by athletes in anxious and excited scenarios. *Psychology of Sports and Exercise*, 5 (2), 201-212.
- 14 **Kar, S. (2013).** Measurement of competition level anxiety of college level athletes by using SCAT. *International Journal of Engineering Science and Innovative Technology (JJESIT)*, vol. 2, issue 3, 367-375.
- 15 **Khodayari, B., Saiiari, A., & Dehghani, Y. (2011).** Comparison Relation between Mental Skills with Sport Anxiety in Sprint and Endurance Runners. *Procedia - Social and Behavioral Sciences*, Volume 30, 2280-2284.
- 16 **Kumar, K. P. (2014).** A Study on Anxiety of Sprinters, Middle & Long distance runners for Effective Performance. *International Journal of Health, Physical Education and Computer Science in Sports* Volume No.15, No.1.98-99.
- 17 **Lawless, F. J. (2013).** A profile and training programme of psychological skills for track and field athletes. <http://hdl.handle.net/10019.1/85644>.
- 18 **Mamassis, G. & Dogains, G. (2004).** The effects of a mental training programme on juniors pre-competitive anxiety, self-confidence, and tennis performance. *Journal of Applied Sports Psychology*, Vol. 16, Issue 2, 118-137.
- 19 **Maruo, Y., Murphy, T. I. & Masaki, H. (2018).** Long-distance runners and sprinters show different performance monitoring—An event-related potential study. <https://doi.org/10.3389/fpsyg.2018.00653>.
- 20 **Nayek, B., & Chatterjee, K. (2013).** Comparative study on pre-competition anxiety between national and state level women athletes. *Journal of Sports and Physical Education*

- (IOSR-JSPE), Volume 1, Issue 2, 33-36. www.iosrjournals.org
- 21 **Neil, R., Mellalieu, S. D. & Hanton, S. (2006).** Psychological Skills Usage and the Competitive Anxiety Response as a Function of Skill Level in Rugby Union. *Sports Science & Medicine*, 5(3): 415–423.
 - 22 **Neil, R., Wilson, K., Mellalieu, S. D., Hanton, S. & Taylor, J. (2012).** Competitive anxiety intensity and interpretation: A two-study investigation into their relationship with performance. *International Journal of Sports and Exercise Psychology*, Vol. 10, Issue 2, 96-111.
 - 23 **Parnabas, V. (2015).** The effect of competitive state anxiety on sports performance. *The International Journal of India Psychology*, vol. 2, special issue, 42-51.
 - 24 **Perry, J. D. & Williams, J. M. (1998).** Relationship of intensity and direction of competitive trait anxiety to skill level and gender in tennis. *Human Kinetics Journal*, Vol. 12, Issue 2, 169-179.
 - 25 **Rathschlag, M. & Memmert, D. (2014).** Self-generated emotions and their influence on sprint performance: An investigation of happiness and anxiety. *Journal of Applied Sports Psychology*, Vol. 27, Issue 2, 186-199.
 - 26 **Reigal, R. E., Juan A. Vazquez-Diz, Juan P. Morillo -Baro, Antonio Hernandez-Mendo & Veronica Morales-Sanchez (2020).** Psychological profile, competitive anxiety, moods and self-efficiency in beach handball players. <https://doi.org/10.3390/ijerph17010241>
 - 27 **Sangari, M., Fotrou, F., & Masrour, F. F. (2012).** Relationship between mental skill and competitive anxiety in female national football players. *World Applied Sciences Journal* 20 (8): 1175-1178, 2012, DOI: 10.5829/idosi.wasj.2012.20.08.2813
 - 28 **Sharma, R. (2013).** Contribution of Psychological Skills in Running Performance of National Level Sprinters. *Indian Journal of Physical Education, Sports and Applied Science* Vol. 3, No. 1, January 2013.
 - 29 **Taylor, J. (1987).** Predicting athletic performance with self- confidence and somatic and cognitive anxiety as a function of motor and physiological requirements in six sports. <https://doi.org/10.1111/j.1467-6494.1987.tb00432.x>
 - 30 **Thomas, O., Hanton, S. & Maynard, I. (2007).** Anxiety response and psychological skill use during the time leading up to competition: Theory to practice I. *Journal of Applied Sports Psychology*, Vol. 19, Issue 4, 379-397.
 - 31 **Wadey, R. & Hanton, S. (2008).** Basic psychological skill usage and competitive anxiety responses. *Research Quarterly for Exercise and Sport*, Vol. 79, Issue 3, 363-373.
 - 32 **Williamson, M. (2019).** The impact of confidence on athletic performance- how the implementation of mental training affect track and field performance. <http://hdl.handle.net/11603/16691>
 - 33 **Znazen, H., Slimani, M. & Miarka, B. (2017).** Mental skills comparison between elite sprint and endurance track and 1 field runners according to their genetic polymorphism: A pilot study.” *J Sports Med Phys Fitness*-6441.